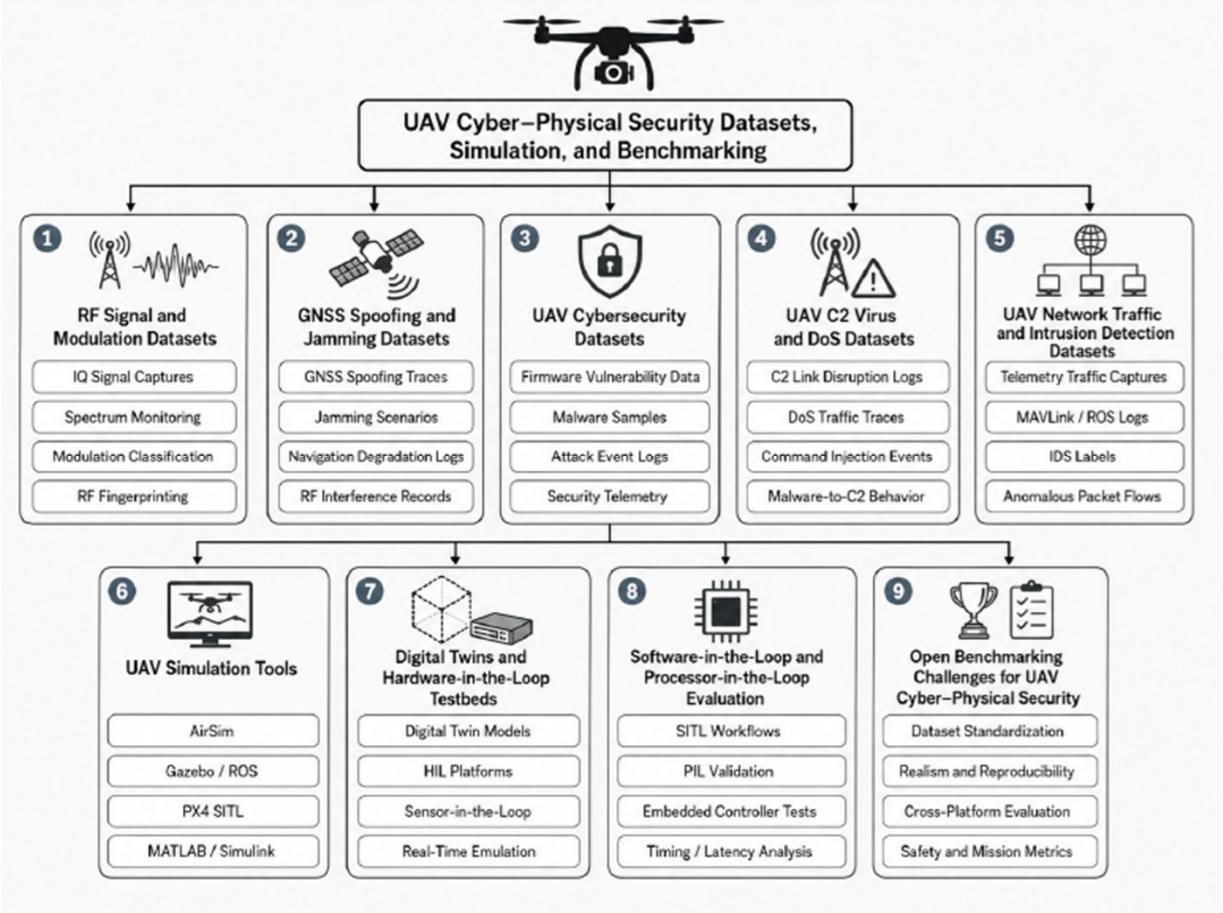
CHAPTER 12 / SUPPLEMENTARY MATERIAL

Cyber–Physical Security and Mission Assurance for Unmanned Aerial Vehicles

Public Datasets, Testbeds, Simulation Environments, and Analysis Tools

Public Datasets, Testbeds, Simulation Environments, and Analysis Tools



RF Signal and Modulation Datasets

Dataset or repository	Access link	Contents
DroneRF	Mendeley Data	RF recordings from three drones in several operating modes, plus background recordings. (data.mendeley.com)
DroneRF Code Repository	GitHub	Companion LabVIEW acquisition tools and MATLAB and Python scripts for drone detection and identification. (GitHub)
RFUAV	Hugging Face	Raw RF recordings and spectrograms for UAV detection and identification. (Datasets at Hugging Face)
DroneRFb-DIR	Science Data Bank	RF recordings for identifying individual drones. (scidb.cn)
KU Leuven Drone RF Dataset	KU Leuven Research Data Repository	Drone RF signals captured in a semi-anechoic chamber. (rdr.kuleuven.be)
Noisy Drone RF Signal Classification	Kaggle	Drone and background samples across multiple signal-to-noise ratios, packaged in a single PyTorch dataset file. (digitalcollection.zhaw.ch)
Noisy Drone RF Signal Classification v2	Kaggle	Individual PyTorch I/Q sample files labeled by class and signal-to-noise ratio. (GitHub)

RF Signal and Modulation Datasets

RadioML 2016.10A	DeepSig dataset page	GNU Radio synthetic benchmark with 11 analog and digital modulation types across multiple signal-to-noise ratios. (DeepSig)
RadioML 2018.01A	DeepSig dataset page	Modulation classification benchmark with 24 analog and digital modulation types, distributed in HDF5. (DeepSig)
WiSig	UCLA CORES Lab downloads	Wi-Fi signal captures for RF fingerprinting and transmitter identification. (CORES Lab - UCLA)
TEXBAT	UT Austin download directory	Texas Spoofing Test Battery with clean GPS recordings and recorded spoofing scenarios. (rnl-data.ae.utexas.edu)
FGI GNSS Jamming and Spoofing Dataset Repository	National Land Survey of Finland	Collection of raw GNSS I/Q recordings for evaluating receiver resilience to jamming and spoofing. (National Land Survey of Finland)

RF Jamming Datasets

Dataset	Access link	Contents
RF Jamming Dataset	GitHub data and code	Experimentally measured wireless spectral scans for detecting RF jamming. (GitHub)
UAV-NIDD	Zenodo CSV files · Figshare raw PCAP files	Benign and attack traffic collected from a physical UAV network testbed. (zenodo.org)
Synthetic UAV Jamming Identification Dataset	ISCTE research paper	Simulated air-to-ground jamming scenarios that vary jammer power and location. I could verify the paper but found no public dataset download. (repositorio.iscte-iul.pt)
Synthetic 5G UAV Attack Dataset	arXiv research paper	Simulated 5G UAV attack scenarios with RSSI and SINR measurements. The paper names UCI as the intended host, but I could not verify a dataset listing there. (arxiv.org)

RF Jamming Datasets

GNSS Interference and Spoofing Dataset	Mendeley Part I · Part II · Part III	Parts I and II contain clean GNSS observations. Part III includes jamming and spoofing recordings, processed data, and scripts. (Mendeley Data)
DroneRF Dataset	Mendeley Data	RF recordings from three drones in different operating modes, plus background recordings. (Mendeley Data)
RFUAV Benchmark Dataset	Hugging Face	Raw RF recordings and spectrograms for UAV detection and identification. (Datasets at Hugging Face)
DroneRFb-DIR Dataset	Science Data Bank	RF recordings for identifying individual non-cooperative drones. (scidb.cn)

GNSS Spoofing and Jamming Datasets

Dataset or resource	Access link	What it contains
Texas Spoofing Test Battery (TEXBAT)	UT Austin download directory	Clean static and dynamic GPS recordings and eight recorded spoofing scenarios. (rnl-data.ae.utexas.edu)
Oak Ridge Spoofing and Interference Test Battery (OAKBAT)	GPS recordings · Galileo recordings	RF recordings from controlled spoofing and interference scenarios for GPS and Galileo. (Oak Ridge National Laboratory)
GNSS Interference and Spoofing Dataset, Mendeley Data	Part I · Part II · Part III	Parts I and II contain clean receiver observations. Part III includes jamming and spoofing measurements, processed data, and scripts. (data.mendeley.com)
FGI GNSS Jamming and Spoofing Dataset Repository (FGI-JSDR)	National Land Survey of Finland	Collection of raw GNSS I/Q recordings, including GPS and Galileo signals under several spoofing conditions. (National Land Survey of Finland)

GNSS Spoofing and Jamming Datasets

Dataset or resource	Access link	What it contains
Jammertest 2024 GNSS Dataset	Zenodo	Field-test data covering jamming, spoofing, meaconing, and combined scenarios. Includes UBX files, RINEX observations, RF monitoring data, and navigation solutions. (Zenodo)
STRIKE3 GNSS Threat Monitoring Data	CORDIS project results and analysis report	Public analysis of interference events collected by an international monitoring network. I could verify the report but found no public download for the underlying event database. (European Commission)
Simulated UAV GNSS Attack Logs	PX4-SITL GPS spoofing dataset and download links	An example dataset for this category. It contains simulated quadcopter sensor logs with attack labels, plus ground-truth position in the raw files. (GitHub)

UAV Cybersecurity Datasets

Attack group	Dataset	Access link	Coverage notes
GPS spoofing	A Dataset for GPS Spoofing Detection on Autonomous Vehicles	IEEE DataPort · GitHub	Simulated UAV GPS spoofing data. (GitHub)
GPS spoofing, GPS jamming, ping DoS	UAV Attack Dataset	IEEE DataPort	Live and simulated flight logs. Ping DoS uses a flood of MAVLink ping messages. (IEEE DataPort)
Deauthentication, DoS, replay, FDI, evil twin	Cyber-Physical Dataset for UAVs Under Normal Operations and Cyberattacks	GitHub CSV and documentation	Four documented attack classes. Deauthentication is the DoS mechanism. The other classes are replay, false data injection (FDI), and evil twin. (github.com)

UAV Cybersecurity Datasets

Attack group	Dataset	Access link	Coverage notes
Malware injection, network intrusion, DoS	Drone-Based Malware Detection (DBMD)	Kaggle	Reviews describe the labels differently. One groups them under malware injection, network intrusion, and DoS. Another reports DDoS, phishing, and malware infection. I could not verify the exact labels in the files. (mdpi.com)
DoS, MitM, spoofing, replay, malware injection	CTGAN Enhanced Dataset for UAV Network Intrusion Detection	IEEE DataPort	A published review lists all five categories for this synthetic benchmark. I could verify the dataset record, while the file-level labels remain unverified. (mdpi.com)

UAV C2 virus and DoS dataset

Dataset	Access link	What it contains
Cyber-Physical Dataset for UAVs Under Normal Operations and Cyberattacks	GitHub CSV and documentation	Network and physical flight features from benign operation and four attacks, including deauthentication DoS, replay, false data injection, and evil twin. (GitHub)
UAV-GCS Intrusion Detection Dataset (UAV-GCS-IDS)	GitHub CSV files · Figshare raw PCAP record	Labeled traffic between a ground control station and a UAV, collected on a physical testbed under benign and attack conditions. The authors list the Figshare record, but I could not verify its download directly. (GitHub)
UAV-NIDD	Zenodo CSV files · Figshare raw PCAP files	Benign and attack traffic from three testbed scenarios involving a compromised UAV, access point, or ground control station. (Zenodo)

UAV C2 virus and DoS dataset

Dataset	Access link	What it contains
MAVLink Message ID Sequence Dataset	GitHub data and code · IEEE DataPort	NumPy arrays of MAVLink message ID sequences from benign and attack runs in a hardware-in-the-loop simulation. (MAVLink Message ID Sequence Dataset)
SHADOW-GCS Attack Dataset	HCRL download page · IEEE DataPort	MAVLink traffic from four UAV–GCS scenarios used to study a relay-based man-in-the-middle attack. (SHADOW-GCS Attack Dataset)
UAVCAN Attack Dataset	HCRL download page · IEEE DataPort	UAVCAN v0 (DroneCAN) traffic from a PX4 testbed, including benign traffic and flooding, fuzzing, and replay scenarios. These links point to the original release. (UAVCAN Attack Dataset)
Tokyo Drone Communication and Security Dataset	Kaggle	Tabular drone-to-drone and drone-to-base-station communication records with features such as timestamps, sender IDs, and encryption status. (kaggle.com)

UAV Network Traffic and Intrusion Detection Datasets

Dataset	Access link	What it contains
UAV-NIDD	Zenodo CSV files · Figshare PCAP files	Benign and attack traffic from a physical UAV testbed, with scenarios involving a compromised UAV, access point, or ground control station. (zenodo.org)
UAV-GCS Intrusion Detection Dataset (UAV-GCS-IDS)	GitHub CSV files · Figshare PCAP record	Labeled traffic between a ground control station and a UAV. The authors link to the raw captures on Figshare, where I could verify the record link but not the download. (github.com)
Cyber-Physical Dataset for UAVs Under Normal Operations and Cyberattacks	GitHub CSV and documentation	Network and physical flight features for benign operation, deauthentication DoS, replay, false data injection, and evil twin attacks. (github.com)

UAV Network Traffic and Intrusion Detection Datasets

Dataset	Access link	What it contains
MAVLink Message ID Sequence Dataset / GUIDE	GitHub data and code · IEEE DataPort	NumPy sequences of MAVLink message IDs from benign and attack runs in a hardware-in-the-loop simulation. (ocslab.hksecurity.net)
MUVIDS False MAVLink Injection Attack Dataset	GUIDE repository with the original MUVIDS data · Original paper	Earlier MAVLink injection data bundled with GUIDE. The two entries therefore share data. (0pt3ryx/GUIDE · GitHub)
UAVIDS-2025	Zenodo CSV	Simulated UAV swarm network flows labeled as normal, blackhole, flooding, Sybil, or wormhole traffic. (Zenodo)
UCI Unmanned Aerial Vehicle Intrusion Detection Dataset	UCI Machine Learning Repository	Encrypted Wi-Fi traffic features labeled by whether the traffic came from a UAV. The labels support UAV identification. (archive.ics.uci.edu)
Synthetic 5G UAV Attack Dataset	arXiv research paper	Simulated 5G UAV attack scenarios using RSSI and SINR. The paper names UCI as the intended host, but I could not verify a public dataset download. (arxiv.org)

UAV Simulation Tools

Network simulator	Access link	Description
ns-3	Official project	Open-source discrete-event simulator for Internet protocols and wired and wireless networks. (nsnam.org)
OMNeT++	Official downloads · INET Framework	Modular C++ discrete-event simulation framework. INET adds network protocol and device models. OMNeT++ is free for academic and nonprofit use. (omnetpp.org)
GloMoSim	University of Virginia documentation · Unofficial source mirror	Legacy UCLA simulator designed for large-scale wireless and mobile networks. I could not verify access to the original UCLA project page. (swarm.cs.virginia.edu)
J-Sim	University of Illinois technical reference	Legacy open-source, Java-based network simulator. The original project domain has changed ownership, and I could not verify an official download. (ideals.illinois.edu)
OPNET Modeler	Riverbed Modeler support and downloads	Network modeling and performance simulation software. Riverbed lists it under the name Riverbed Modeler. (support.riverbed.com)
NetSim (Tetcos)	Official product page · Documentation	Network simulator with models for technologies including 5G and satellite communications. (Network, Satellite & Cyber-Attack Simulation)
QualNet	Keysight technical overview	Network simulation platform for evaluating communication performance under different operating conditions. The link provides vendor documentation. (keysight.com)

UAV Simulation Tools

Tool	Type	Official link	What it does
AirSim	Robotics simulator	Microsoft GitHub repository · Documentation	Simulates drones and other vehicles in Unreal Engine, with virtual sensors and realistic rendering. Microsoft has ended updates to the original AirSim. (AirSim)
FlightGear	Flight simulator	Official downloads	Free, open-source simulator for aircraft flight dynamics, scenery, and flight operations. (FlightGear)
MATLAB UAV Toolbox	Simulation and development toolbox	MathWorks product page · Documentation	Supports UAV scenario simulation, flight-controller design, mission planning, and autonomy testing in MATLAB and Simulink. (mathworks.com)
jMAVSim	Multirotor simulator	PX4 setup guide	Lightweight Java simulator for flying a virtual quadcopter controlled by PX4 in SITL. (PX4 User Guide)

UAV Simulation Tools

Tool	Type	Official link	What it does
AirSim	Robotics simulator	Microsoft GitHub repository · Documentation	Simulates drones and other vehicles in Unreal Engine, with virtual sensors and realistic rendering. Microsoft has ended updates to the original AirSim. (AirSim)
Gazebo	Robotics physics simulator	Gazebo Sim documentation	Simulates robot dynamics, environments, and sensors. The link points to Gazebo Sim. Gazebo Classic reached end of life in January 2025. (Gazebo documentation)
PX4	Autopilot flight stack	PX4 Autopilot · Simulation guide	Runs the flight-control software that commands the simulated vehicle. Supports software-in-the-loop and hardware-in-the-loop testing. (PX4 Guide (main))
SITL	Software-in-the-loop testing mode	PX4 simulation guide · ArduPilot SITL guide	Runs autopilot code on a computer with simulated vehicle dynamics and sensor inputs, without a physical flight controller. (PX4 Guide (main))
X-Plane	Commercial flight simulator	Laminar Research official site	Desktop aircraft simulator that uses blade-element theory to model flight dynamics. (The world's most advanced flight simulator.)

UAV Simulation Tools

Tool	Project or download link	What it connects
ROS-NetSim	GitHub repository	Connects ROS applications to robotics and network simulators through a common interface. The framework supports different simulator backends. (GitHub)
AVENS	GitHub repository · OMNeT++ project listing	Couples X-Plane flight simulation with OMNeT++/INET network simulation to study UAV coverage and connectivity. (GitHub)
CUSCUS	UTC GitLab backend · Heudiasyc project page	Couples FL-AIR for UAV dynamics and flight control with ns-3 for network simulation. The GitLab link is for the backend. (Heudiasyc)
CORNET	Original repository · Successor framework	The original research integrates ROS/Gazebo with ns-3 for multi-UAV experiments. The successor supports ROS 2/Gazebo with ns-3 or Mininet-WiFi. (CNI)

UAV Simulation Tools

Tool	Project or download link	What it connects
FlyNetSim	GitHub repository	Couples ArduPilot SITL with ns-3 through publish–subscribe middleware to study UAV flight and network behavior together. (GitHub)
RoboNetSim	Author's project and download page	Synchronizes multi-robot motion with wireless network simulation. The downloadable package integrates ARGoS or Player/Stage with ns-3. (giannidicaro.com)
SUMO-based tools	Veins · Eclipse MOSAIC · SUMO TraCI documentation	Use SUMO for road traffic and vehicle mobility. Veins couples SUMO with OMNeT++ for vehicular communications, while MOSAIC supports broader connected-mobility co-simulation. (veins.car2x.org)

UAV Simulation Tools

Ground control station	Official link	Main use
Mission Planner	ArduPilot download and installation	Graphical station for configuring ArduPilot, planning waypoint missions, and monitoring live telemetry. Designed primarily for Windows. (Mission Planner documentation)
QGroundControl	Official download and installation guide	Cross-platform graphical station for vehicle setup, mission planning, and flight monitoring with PX4 and ArduPilot. (QGC Guide (Stable_V5.1))
UgCS	SPH Engineering downloads	Professional drone mission planning for mapping, surveying, and terrain-following routes. (sphengineering.com)
MAVProxy	ArduPilot installation guide	Command-line MAVLink ground station for vehicle control and forwarding telemetry to other ground stations over UDP. (MAVProxy documentation)
APM Planner 2	GitHub releases · ArduPilot documentation	Open-source graphical ground station for MAVLink autopilots, including APM and PX4/Pixhawk. (GitHub)